Rethinking Utility in Differentially Private Visualization: A Channel-Centric Perspective

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Abstract

Differential privacy (DP) is increasingly adopted to protect sensitive data in visualization, but understanding how privacy-induced noise affects visual representations remains fragmented. Prior work has examined the impact of DP noise on specific chart types, yet these findings are typically tied to particular chart types and evaluation settings, making it difficult to generalize across contexts. In parallel, visualization research has long established that visual channels, the visual properties of marks that encode data attributes, differ in perceptual effectiveness under noise-free conditions. However, these principles have not been systematically revisited under DP, where noise is deliberately introduced into the data to protect privacy.

In this paper, we present a channel-centric perspective to reframe prior empirical, computational, and theoretical work. Our synthesis draws on prior findings to motivate a channel-read-mechanism hypothesis: the utility of DP visualizations may depend on the task-relevant reads made salient by visual channel configurations and DP mechanism choice jointly. Moreover, this perspective highlights the role of channel choice throughout the privacy-preserving visualization pipeline, from disclosure framing and DP algorithm design to utility evaluation and interpretation of noisy outputs. The synthesis also identifies gaps in the existing evaluation literature and highlights research opportunities for channel-aware DP visualization.

Index Terms: Differential privacy, Visual channels, Privacy-preserving visualization, Visualization utility.

1 Introduction

Differential privacy (DP) has become a widely adopted paradigm for protecting sensitive data in statistical analysis and visualization [4]. It ensures that the output of an analysis remains statistically indistinguishable when any single individual's data is added or removed [5]. More formally, given any neighboring databases $\mathscr{D}$ and $\mathscr{D}'$ that differ by at most one record, a randomized mechanism $\mathscr{A}$ satisfies **(ε, δ)-DP** if for any subset $S \subset range(\mathscr{A})$,

$$\Pr[\mathscr{A}(\mathrm{D}) \in \mathrm{S}] \leq \mathrm{e}^{\varepsilon} \cdot \Pr[\mathscr{A}(\mathrm{D}') \in \mathrm{S}] + \delta. \quad (1)$$

The parameters $\varepsilon \geq 0$ and $\delta \geq 0$ specify the degree of privacy provided by the mechanism. Smaller ε and δ values indicate stronger privacy protection, and vice versa. For aggregate queries, DP guarantees can be achieved by adding random noise into query results, such as through the Laplace mechanism and the Gaussian mechanism [5]. The noise magnitude depends on the query's sensitivity and privacy parameters (smaller ε and δ indicate stronger privacy and may yield larger noise). Most existing studies on DP visualization [9, 18, 27] adopt the ε-DP model (i.e., setting $\delta = 0$). They also adopt algorithmic approaches such as data aggregation and transformation to preserve utility while providing DP guarantees.

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Building on Saket et al.'s [22] task-based evaluation of chart effectiveness under noise-free conditions, Zhang et al. [27] revisited chart-task combinations under DP (via Laplace mechanism) and found that utility degradation varied substantially across chart-task pairs, suggesting that the privacy mechanism does not affect visualizations uniformly. Subsequent work in DP visualization has expanded the design space by showing that choices such as rendering (e.g., bin size and geometry), DP mechanism, and disclosure tactics also influence visualization utility [9, 14, 18, 20]. Together, these findings motivate a need to connect evidence across studies and examine it through a common lens.

Visualization research has long established that visual channels (e.g., position, length, angle, and color/intensity), the perceptual dimensions used to encode data attributes into visual marks [15], differ in perceptual effectiveness [3]. Because chart types are constructed from combinations of visual channels, a channel-centric perspective provides a natural lens for connecting findings across chart types and reasoning about their visual utility under DP. However, this perspective has not yet been used to synthesize findings across DP visualization studies.

In this paper, we synthesize prior empirical, computational, and theoretical work on DP visualization [2, 9, 14, 17, 18, 19, 20, 21, 24, 25, 26, 27, 29] through a channel-centric lens. We identified relevant work using Zhang et al. [27], Panavas et al.'s study [18], and the survey of Bhattacharjee et al. [1] as starting points for backward and forward citation snowballing, supplemented by independent keyword searches (e.g., *differential privacy*, *visual utility*). We prioritized studies examining visualization utility under DP.

When viewed collectively through this channel-centric lens, prior studies motivate the channel-read-mechanism hypothesis for DP visualization utility: utility may partially depend on the interaction between the task-relevant reads made salient by visual channel configurations and the information preserved by the adopted DP mechanisms. We further show that channel choice matters across the DP-visualization pipeline, from disclosure framing and DP algorithm design to utility evaluation and interpretation.

This paper makes three contributions: (1) a channel-centric synthesis of prior empirical, computational, and theoretical work on DP visualization; (2) the channel-read-mechanism hypothesis, which posits that DP visualization utility depends in part on the interaction among visual channel configurations, the task-relevant reads they make salient, and DP mechanism choice; and (3) gaps and research opportunities for channel-aware evaluation and design in DP visualization.

2 Evaluation Context: A Utility Hierarchy

Evaluating the utility of privacy-preserving visualizations is itself a multi-faceted problem. Bhattacharjee et al. [1] identify a fundamental lack of consensus on how to quantify this utility. We characterize three complementary evaluation lenses used in prior work, organized along a hierarchy from low-cost automated methods to high-cost human evaluation.

Visual utility measures how closely a private visualization preserves features relevant for downstream tasks, typically through automated metrics: image-similarity measures or task-specific distri-

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bution scores. Panavas et al. [18] benchmarked automated metrics against expert-perceived visual quality for differentially private binned scatterplots; MS-SSIM (multi-scale structural similarity) emerged as the strongest predictor, while APQE (average per-query error) and Earth Mover's Distance (EMD), two standard DP utility metrics, correlated negatively with perceived visual quality. Ryan et al. [21] showed through user studies that Pixel Approximate Entropy (PAE) correlates with perceived visual complexity in line charts, though not under DP noise. Zhang et al. [27] illustrate the latter category, introducing Peakedness, Anomaly, and Clusteredness scores tailored to three analytical tasks (Find Maximum, Find Anomalies, Cluster) to compare DP algorithms by how well their noisy output preserves the distribution features each task requires.

Task-based utility measures whether human viewers can correctly perform analytical tasks on noisy charts. Zhang et al. [27] proposed a dichotomous assessment model that distinguishes *perceptual accuracy* (did the user correctly read the noisy chart?) from *perturbation accuracy* (does the noisy answer match the true answer?), enabling targeted diagnosis of where utility breaks down. Task-based evaluation is the most direct measure of utility but also the most expensive, requiring human participants for each combination of conditions.

Perceptual utility operates along a distinct axis: it measures whether an authorized viewer can read the chart while unauthorized viewers cannot, addressing physical observation threats rather than inference from released outputs [2, 28].

Notably, the automated metrics used at the lower end of this hierarchy already differ by chart type and associated channel: MS-SSIM for binned scatterplots [18], PAE for line charts [21]. Whether this pattern reflects a deeper need for channel-specific evaluation is a question we return to in Section 4.

3 A Channel-Centric Synthesis of DP Visualization Utility

We organize our cross-study synthesis into two themes. Each draws on multiple studies whose connections become apparent through the lens of visual channels.

3.1 Observations of Utility Variation under DP

Existing studies have reported differences in visualization utility under DP across a variety of visualization settings. The evidence spans comparisons across chart types as well as within-channel design choices, including discretization, DP mechanisms, component-specific estimation, and rendering geometry.

Chart-task robustness. Zhang et al. [27] evaluated how the Laplace mechanism affects different chart-task combinations, using a user study that measured perceptual accuracy against non-private counterparts. Their results showed that perceptual accuracy degradation varied across chart-task pairs. For example, pie charts exhibited relatively small degradation for Filter, barplots degraded the most for Find Maximum, and line charts experienced the largest degradation for Retrieve Value, Filter, and Determine Range among the evaluated chart types. Panavas et al. [18] observed a related task-level pattern for binned scatterplots, generated with 5 DP mechanisms: Cluster retained the highest visual utility under DP, whereas Distribution retained the lowest.

Discretization parameters. Panavas et al. showed that, holding the privacy level and DP mechanism constant, coarser bins (32×32) retain significantly more visual utility than finer bins (64×64) [18]. Ghazi et al. [9] suggested a possible explanation: finer grids split the same data across more bins, reducing per-bin signal while leaving DP noise magnitude largely unchanged. This lowers the signal-to-noise ratio of spatial patterns, suggesting that the bin-size effect reflects a broader discretization challenge in DP spatial visualization.

DP mechanisms. Holding the visualization fixed, prior work has shown that the choice of DP mechanism can substantially affect visualization utility. Zhang et al. [27] showed through metric-based simulations that, for histogram-based tasks, DAWA slightly outperformed Laplace for Find Anomalies, whereas Laplace better preserved task answers for Find Maximum and Cluster. For color/intensity-encoded binned scatterplots, Panavas et al. [18] used expert-rated visual utility and metric analysis to compare mechanisms, showing that different mechanisms yielded different levels of visual utility across analytical tasks. Together, these studies suggest that, even when the visual channel configuration is held fixed, different DP mechanisms can preserve different task-relevant structures, leading to different levels of visualization utility.

Component-specific estimation. Ramsay and Diaz-Rodriguez [20] showed that, although the box and whiskers of a boxplot both use the position channel, they summarize different statistical quantities: the box represents the median and quartiles, whereas the whiskers depend on extreme quantiles. Their composite DPBoxplot therefore applies different DP estimators to those components, demonstrating that component-aware estimation can reduce utility loss relative to using a single estimator throughout the chart.

Rendering geometry. Ryan et al. [21] showed that line chart geometry, independent of the underlying data, substantially shifts visual complexity, as measured by PAE: doubling chart width reduces PAE by more than half, while doubling height increases it more than twofold. Combined with their experimental finding that shape identification accuracy drops sharply once PAE exceeds a threshold, this implies that aspect ratio may mediate utility for private line charts: a wider rendering may lower perceived visual complexity, helping keep the chart below the threshold at which trends become unrecoverable. Thaker et al. [26] made a complementary observation for bar/histogram length encodings: at sufficiently large data scales, DP perturbations can fall below the screen's pixel resolution, making a noisy bar visually indistinguishable from its non-private counterpart. Together with Ryan et al.'s PAE result, this shows that rendering geometry can act as a within-chart design lever.

The studies reviewed above collectively suggest that tasks, visual channels, task-relevant quantities, and DP mechanisms jointly shape visualization utility. We use *read* to refer to the task-relevant quantity a viewer seeks to recover from a visualization. Table 1 summarizes these observations together with our interpretation of task-relevant channel configurations and task-relevant reads made salient by them. Note that charts are often multi-channel encodings and we list the task-relevant channel(s) following canonical channel vocabulary [3, 15]. Piecharts are represented by arc length, central angle, and segment area jointly because prior studies suggest that all three cues can affect proportional reading [6, 23]. Collectively, these observations motivate the hypothesis that visualization utility depends in part on the interaction among the visual channel configuration, the task-relevant read, and the adopted DP mechanism. We caution that these observations are derived primarily from a small number of studies. Section 4.1 develops this hypothesis.

3.2 Channel Considerations Across the DP-Visualization Pipeline

The channel-centric view shapes practice upstream and downstream of visualization selection. We trace its implications at four points along the DP-visualization pipeline: how disclosure is framed in design, how DP mechanisms are tuned to preserve specific structures, how utility is measured, and how analysts interpret noisy output. At each point, the channel acts not as a passive rendering decision but as an active factor shaping what privacy-preserving visualization can deliver.

Disclosure and channel as paired design choices. Mehta et al. [14] contributed a theoretical framework that treats channel as

Table 1: Observations motivating the channel–read–mechanism hypothesis. PP = post-processing; GT = Geometric Truncated; DAWA/MWEM/AGrid/AHP = DP mechanisms used in [18, 27].

Source	Chart	Task	Task-relevant channel(s)	Task-relevant read	DP mechanism	Key observation
Zhang et al. [27]	Lineplot	Characterize Distribution	Connected position	Trend/shape	Laplace + PP	Largest degradation among evaluated charts.
Zhang et al. [27]	Barplot	Find Maximum	Length	Absolute magnitude	Laplace + PP	Greater degradation than other evaluated chart types for this task.
Zhang et al. [27]	Piechart	Filter	Arc length, central angle, and segment area jointly	Part–whole	Laplace + PP	Relatively small degradation.
Zhang et al. [27]	Histogram	Max; Anomaly; Cluster	Length	Peak; Anomaly; Cluster	Laplace; DAWA; MWEM	DAWA better for Anomaly at a lower privacy budget; Laplace better for Max and Cluster.
Panavas et al. [18]	Binned scatterplot	Cluster; Correlation; Distribution	Color/intensity	Dense regions; Association pattern; Overall distribution	Laplace; GT; DAWA; AGrid; AHP	Clustering retained the highest utility; Distribution the lowest. Coarser binning retained higher utility than finer binning.
Ramsay & Diaz-Rodriguez [20]	Boxplot	Center/spread; Extremes; Outliers	Position along a common scale	Median/quartiles; Min/max; Outlier values	Laplace; unbounded; JointExp	Different estimators improved different components.

design-relevant in privacy-preserving visualization. They distinguished *disclosure tactics* (data operations that change what information is available) from *emphasis tactics* (rendering choices like mark types, scales, and layouts). Their Fred scenario illustrates that disclosure tactics and channels are not chosen independently: a biostatistician sharing patient clusters while hiding individuals deliberately pairs the *smoothing* tactic with *contour shape*, because that combination preserves cluster visibility while suppressing individual-level detail. They acknowledged that emphasis-tactic choices "tend to be concomitant to choices about disclosure" but left emphasis tactics out of their analytical scope. From a channel-centric perspective, this omission points to an important design gap: under DP noise, disclosure tactics and channel choices may jointly influence which aspects of utility are preserved.

DP mechanism, channel, and task as joint design choices. Ghazi et al. [9] designed a DP mechanism specifically for heatmap visualization, with EMD as the loss function the mechanism targets, chosen for its spatial-proximity properties on color/intensity-encoded heatmaps. This illustrates DP mechanism design being tailored to a target visualization rather than treated as an independent, upstream choice. Panavas et al. [18] observed a related coupling within a single channel: DAWA [12], a data-dependent mechanism that groups nearby cells into near-uniform regions, reinforces the dense-region signal that cluster identification depends on while smoothing away the subtle density variations that distribution characterization requires. Their results showed that, under DAWA, cluster-identification tasks retained higher visual utility than distribution-characterization tasks. Whether mechanisms tuned for one channel–task pairing transfer to others remains an open question that requires empirical evaluation. The evidence so far suggests that selecting a DP mechanism without considering both the visual channel and the analytical task risks discarding what the visualization is meant to communicate.

Evaluation metrics need empirical validation, not just theoretical channel-fit. The choice of evaluation metric is itself a design decision that interacts with channels and tasks. Panavas et al. [18] empirically benchmarked five automated metrics (MS-SSIM, APQE, EMD, KS-Test, and Scagnostics) against expert ratings of visual utility across multiple heatmap-style binned scatterplots analysis tasks. MS-SSIM correlated most strongly with the expert ratings, whereas APQE, despite being a standard DP statistical metric, and EMD both correlated negatively. Theoretical arguments for a metric's channel-fitness do not automatically translate to predictive power for perceived utility. This is consistent with Panavas et al.'s [19] practitioner-interview finding that metrics tailored to particular scenarios are often more useful than general metrics.

The interpretation bottleneck. Nanayakkara et al. [17] found that analysts using DP outputs lost roughly nine times as much performance to misreading noisy results as to allocating the privacy budget poorly. ViP [16] illustrated the leverage of a visualization design choice: it discretizes the continuous Laplace noise distribution into a quantile dotplot, turning probability-mass reading into counting dots rather than estimating area under a curve. A controlled study with 16 practitioners showed analysts read CDF probabilities more accurately from the dotplot than from a numerical-error baseline. This substitution favors count-based judgment over continuous-magnitude reading. If interpretation is a primary bottleneck, visualization design choices are where direct leverage lies.

4 Discussion

4.1 The Channel-Read-Mechanism Hypothesis

The observations in Table 1 motivate the **Channel-Read-Mechanism Hypothesis**: The utility of a privacy-preserving visualization may partially depend on the interaction among the visual channel configuration, the task-relevant read it makes salient, and the adopted DP mechanism. Because different DP mechanisms may preserve different properties of the data, visualization utility depends in part on how well the adopted mechanism preserves the information required by the task-relevant read.

For lineplots, adopting the Laplace mechanism [5] does not guarantee preservation of ordering between adjacent data points, and may therefore alter upward or downward trends. Such disruptions can introduce contour discontinuities, increasing visual complexity as measured by PAE [21]. Prior research shows that post-processing the results of the Laplace mechanism (e.g., to enforce non-negativity) may introduce bias and different error variances among data points [8], which may explain the difficulty with retrieving magnitudes accurately in DP barplots. Laplace better preserved task answers for Find Maximum and Cluster in Zhang et al.'s histogram results [27], tasks that involve identifying dominant or dense groups. This pattern is consistent with prior work on frequent itemset mining and web mining, where Laplace-based mechanisms have been shown to preserve large counts or frequent patterns [7, 11, 13].

The proposed hypothesis also suggests implications for within-channel design choices and a broader range of DP mechanisms. Coarsening bin resolution or performing hierarchical decomposition [9, 18, 26] could better preserve aggregate density and thus retain better visual utility. Ramsay and Diaz-Rodriguez's [20] box-

plot decomposition shows that private quantile estimators may perform differently towards inner quantiles and extreme quantiles. Incorporating specific pattern constraints in the DP mechanism may improve the visualization utility in pattern retention [29].

However, existing evidence is limited. Existing studies have not yet fully isolated the channel-read-mechanism interaction from confounds such as label design, visual complexity, or task familiarity. For example, Saket et al. [22] noted that pie charts may have benefited from colored slices with text labels. Thus, the hypothesis remains a conjecture awaiting controlled studies.

4.2 Gaps

Visual utility metrics are channel–task specific and largely unvalidated. As noted in Section 2, the automated metrics already in use differ by visualization: MS-SSIM for binned scatterplots [18], PAE for line charts [21]. Other channel–task combinations, including those involving angle, length, and slope, lack perceptually validated automated metrics. Theoretical arguments for a metric's fitness do not automatically translate to predictive power for perceived utility, as Panavas et al. [18] demonstrate empirically (Section 3). Validating which metrics effectively track perceived utility for specific channel–task pairs therefore requires empirical work that remains largely undone for most visualization types.

Task-based evidence rests on narrow conditions. Zhang et al.'s [27] task-based user study, a primary source of observations about chart–task variation under DP, varied chart type, task, and privacy level, but held fixed several potentially consequential dimensions: within-channel design parameters such as bin resolution and chart geometry, data properties such as scale and distribution, and the choice of DP algorithm (Laplace only). Whether the documented utility-degradation patterns hold under different conditions remains unvalidated in task-based user studies. Length encoding has been studied at the statistical level by Hay et al. [10] across scale, distribution, domain size, and DP algorithm; whether their statistical findings translate into task or visual utility for barplots remains uncharacterized. Visualization types beyond basic charts, such as networks, treemaps, and parallel coordinates, are largely unexamined under DP, with Tao et al.'s [25] brain connectivity work a rare exception.

Perceptual utility has been studied largely independently of DP. BAIT [2], a visual-illusion-based perceptual privacy approach, demonstrated uniformly high protection across chart types for shoulder surfing but found channels differ in vulnerability: angle and slope encodings are most "glanceable." Whether channel effects derived under perceptual privacy align with or diverge from observations under DP has not been examined, leaving the channel-level relationship between the two privacy regimes open.

4.3 Opportunities

A channel–task-specific metrics suite. The methodology Panavas et al. [18] used to validate MS-SSIM (collecting expert visual utility ratings on a corpus of DP charts, then benchmarking candidate automated metrics against those ratings) may transfer directly to other channel–task pairs. Developing such a suite would let practitioners compare visualizations at scale during design, rather than after deployment through expensive user studies.

Evaluation across channels and tasks at matched ε. Extending Zhang et al.'s [27] user study design to cover underrepresented channel–task combinations, while varying data scale, distribution, within-encoding design choices, and DP mechanisms, would provide a more systematic evaluation of visualization utility under DP. Such studies would foster a deeper understanding of how tasks, channels, task-relevant reads, and DP mechanisms interact, and could inform new designs that better preserve utility.

Cross-channel evaluation of structure-aware DP mechanisms. Panavas et al. [18] showed that DP mechanism choice substantially affects visual utility within a fixed channel configuration, and that a single mechanism's fit can vary by task on the same channel (Section 3). Whether such differences extend across channels and tasks, and whether mechanisms purpose-built for specific channels or tasks (e.g., [9, 20, 24]) transfer to others, has not been tested under matched conditions.

Multi-channel interactions. Most real-world visualizations, including many basic charts, rely on multiple visual channels. More complex visualizations, such as treemaps, parallel coordinates, and node-link diagrams, often involve even richer combinations of channels. How channels interact under DP and whether noise on one channel compounds or compensates for noise on another remain largely open. Such interactions may also complicate mechanism selection: a mechanism optimal for a chart's primary channel and task-relevant read may degrade secondary channels or reads. It is possible that DP mechanism design for multi-channel encodings requires joint optimization across channels and reads.

Channel behavior across privacy threat models. Whether BAIT's [2] channel-level vulnerability findings interact with DP-induced degradation is unexplored. A study that varies observation threat (BAIT-style), DP noise, visual channel, and analytical task would clarify whether the two privacy regimes reinforce or counteract each other for particular channel–task combinations.

4.4 Limitations

The utility-degradation patterns identified in prior work are specific to the chart types, channel configurations, tasks, DP mechanisms, privacy parameters, datasets, and study methods used in those studies. The channel–read–mechanism hypothesis we propose, while consistent with available evidence, has not been tested in a controlled experiment that isolates channel/read properties from confounds.

Despite these limitations, our synthesis reveals patterns that are less visible when individual studies are considered in isolation. It shifts attention from "how much noise can we add?" to "how should we represent the noisy data, and how does that choice shape disclosure framing, mechanism selection, and interpretation?" This is a question visualization researchers are uniquely positioned to answer. Panavas et al.'s [19] practitioner-interviews reveal that practitioners select chart types for different pipeline stages without an established framework; their interviewees express a desire for empirically grounded design guidance. Developing empirically grounded, channel–task-aware design principles is therefore essential for making DP visualizations both private and usable. Such guidance is essential if DP visualization is to move from isolated chart-specific findings toward reusable design principles, and may also inspire new DP mechanisms that better preserve visualization utility.

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